Christian Bianchi

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Education

Sapienza University of Rome — Rome, Italy
M.Sc. in Computer Science, Honours Programme; GPA 28.57/30 — *Expected Jul 2027*

- Thesis: reinforcement learning for humanoid control; advisors Georg Martius and Fabio Galasso.

Sapienza University of Rome — Rome, Italy
B.Sc. in Applied Computer Science and Artificial Intelligence — *Oct 2024*

- Thesis: *Calibrating Neural Networks via Radius Regularization* (advisor Indro Spinelli); reduced expected calibration error on CIFAR-100 by 57% at unchanged accuracy. Completed while working full-time.

Publications

C. Bianchi, M. Amlal, A. Palma, M. Gioia, E. Ensoli, A. Sampieri, A. Roberti, F. Galasso, and L. Franco. **Long-Horizon Compositional Manipulation via Meta-Network Dynamic Adaptation.** *Under review*, 2026.

C. Bianchi, S. Yousefi, A. Sampieri, A. Roberti, L. Rigazio, F. Galasso, and L. Franco. **Robotic Policy Adaptation via Weight-Space Meta-Learning.** *CoRL Workshop on Compositional Generalization*, 2026. [paper | page]

L. Sigillo, **C. Bianchi**, A. Uncini, and D. Comminiello. **Quaternion Wavelet-Conditioned Diffusion Models for Image Super-Resolution.** In *International Joint Conference on Neural Networks (IJCNN)*, 2025. **Oral.** [paper | code]

Research Experience

Autonomous Learning Group, University of Tübingen — Sep 2026 – Present
Visiting Student Researcher (M.Sc. Thesis), with Georg Martius and Fabio Galasso — *Tübingen, Germany*

- Studying weight-space steering of a pretrained humanoid motion tracker after training, without retraining it, along trade-offs such as accuracy versus smoothness (AMOR-style control through the weights of an existing controller).

ItalAI & PINlab, Sapienza University of Rome — Jun 2025 – Present
R&D Scientist, with Fabio Galasso and Luca Franco — *Rome, Italy*

- Raised held-out LIBERO-Spatial success from 19% (fine-tuned $\pi_{0.5}$) and 28% (LoRA-Gen) to 40% by designing **WIZARD**, a meta-network that generates LoRA adapters for a frozen VLA policy from one unlabeled demonstration; 17% to 37% on a real Franka Emika Panda.
- Doubled held-out LIBERO-10 success (0.25 to 0.50) and reached the best CALVIN ABC→D score (4.58) by designing **SPELL**, which generates skill-level adapters and switches them with learned termination heads; 0.31 to 0.56 on a real Franka.
- Led both projects as first author from problem formulation to real-robot evaluation and writing, coordinating experiments across 2 Ph.D. and 1 M.Sc. researchers.
- Co-supervised an M.Sc. thesis on multimodal game agents and a B.Sc. thesis on 3D hair generation.

ISPAMM Lab, Sapienza University of Rome — Dec 2024 – May 2025
Student Researcher, with Danilo Comminiello and Luigi Sigillo — *Rome, Italy*

- Improved PSNR on DRealSR from 27.87 to 29.69 over the strongest diffusion baseline by conditioning latent diffusion on quaternion wavelet embeddings (ResQu); presented as an oral at IJCNN 2025.

Industry Experience

F1 Consulting Services — Mar 2025 – Jun 2025
Generative AI Engineer — *Rome, Italy*

- Contributed to the Aeroporti di Roma Virtual Assistant, a multi-agent Amazon Bedrock assistant on WhatsApp and the web for an airport system serving about 50M passengers a year, building its voice pipeline (Whisper on SageMaker, Polly). [demo | article]
- Contributed to a cloud IVR modernization for Cassa Depositi e Prestiti (Amazon Connect, Lex).

HCL Software — May 2022 – Mar 2025
Software Engineer II — *Rome, Italy*

- Built dense-vector retrieval infrastructure with approximate nearest-neighbor indexing, improving indexing capacity, memory use, and query latency; one of four HCL Software Maverick Award recipients worldwide (2024).

Research Projects

Privileged Context and Self-Distillation for Multimodal Game Agents | *Qwen3-VL*, *GameWorld* Ongoing

- Raised Qwen3-VL-30B overall success from 12.4% to 14.9% and progress from 30.6% to 35.8%, above the 8× larger Qwen3-VL-235B (13.5%, 30.8%), by evolving game-knowledge instructions with GEPA at fixed weights.
- Next: isolating which in-context information (memory, evolved instructions, oracle game state) makes the agent succeed, then internalizing it into the weights, by on-policy self-distillation (OPSD).

Tool-Use Adaptation with Hypernetworks | *PyTorch*, *AppWorld* 2026

- Generating low-rank adapters that teach frozen language models new tool-use skills from demonstrations; evaluated on synthetic tasks and AppWorld APIs.

Machine Unlearning for 3D Human Motion Generation | *PyTorch* [code] 2026

- Adapted three unlearning methods (ESD, LoRA, UCE) to a text-conditioned motion diffusion model and evaluated targeted forgetting against preservation of unrelated motions.

Physics of Flow Matching: Steering FLUX Velocity Fields | *PyTorch*, *FLUX.1* 2026

- Decomposed the learned velocity field of a flow-matching image model via Helmholtz–Hodge into irrotational and divergence-free components, recombining them across prompts and steering sampling with Lie-bracket corrections.

Cross-Modal Retrieval with Hard-Negative Contrastive Learning | *PyTorch* 2025

- Mapped text embeddings into an image embedding space with a transformer trained on a symmetric InfoNCE loss restricted to the top-k hardest negatives, with target-space mixup and Optuna search over loss and architecture.

Presentations and Service

Reviewer, International Joint Conference on Neural Networks (IJCNN) 2027

Poster, *Robotic Policy Adaptation via Weight-Space Meta-Learning*, CoRL Workshop, Austin 2026

Poster, *Robotic Policy Adaptation via Weight-Space Meta-Learning*, ICVSS, Sicily 2026

Oral presentation, *Quaternion Wavelet-Conditioned Diffusion Models for Image Super-Resolution*, IJCNN, Rome 2025

Student volunteer, SIGGRAPH, Vancouver 2025

Student volunteer, International Joint Conference on Neural Networks (IJCNN), Rome 2025

Honors and Awards

Selected participant, Machine Learning Summer School (MLSS), Tübingen 2026

Selected participant, International Computer Vision Summer School (ICVSS), Sicily 2026

Honours Programme, M.Sc. in Computer Science, Sapienza University of Rome 2026

Selected mentee, LeadTheFuture, Class of 2027 2026

Full scholarship for study abroad, sole recipient among 2,000+ applicants 2019

Technical Skills

Languages: Python, C++, C, Java, SQL, Bash
ML: PyTorch, JAX, Hugging Face Transformers, vLLM; RL, imitation learning, VLA models, diffusion, LoRA
Robotics: LIBERO, CALVIN, MetaWorld, Isaac Sim, MuJoCo; Franka Emika Panda, Unitree G1
Infrastructure: Slurm, distributed training, Docker, Git, Linux, AWS